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Linear algebra, winter 2021, CS.

Problema 4

1. Consider the linear manifolds

$$L = \{(x, y, z, t) \in \mathbb{R}^4 | x - y + z - t = 1\},$$

and

$$K = \{(x, y, z, t) \in \mathbb{R}^4 | x - y + z - t = 0, x + y = 1\}.$$

1. Find the distance between L and K , $d(L, K)$;
2. Find the dimension of L and K .

2. Consider the linear manifolds

$$L = \{(x, y, z, t) \in \mathbb{R}^4 | x - y + z - t = 1\},$$

and

$$G = \{(x, y, z, t) \in \mathbb{R}^4 | a_1x + b_1y + c_1z + d_1t = e_1, a_2x + b_2y + c_2z + d_2t = e_2\}.$$

Give some sufficient conditions, by using the matrix of the system which consists of the equations from both manifolds for $d(L, G) = 0$ and $d(L, G) \neq 0$ respectively. Try to explain the conditions in the properties of the linear manifolds.

Note. In the file submitted please write computations, explanations and everything needed for a better understanding of the problem.